THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sudden drop in pressure at the centre of a tornado can cause houses to: 1

(a) stay intact **(b)** get magnetised **(c)** become cold **(d)** burst outwards

2. Assertion (A): Atmospheric pressure decreases with height. Reason (R): The amount of air above decreases as we go up. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The air pressure in a car tyre is measured at 200 kPa. The force on 0.01 square metre of the tyre wall is: 1

(a) 2000 N **(b)** 20000 N **(c)** 200 N **(d)** 20 N

4. The windward side of a mountain has more rainfall because: 1

(a) moist air rises and cools there **(b)** air sinks there
(c) it has no clouds **(d)** it is nearer to the Sun

5. Assertion (A): The water jet from the lowest hole of a bottle goes farthest. Reason (R): Pressure of liquid increases with depth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. In a barometer the reading at a hill station is lower than at sea level because: 1

(a) the barometer is faulty **(b)** hill air is colder
(c) hills attract less **(d)** there is less air above the hill station

7. A person climbing Mount Everest carries oxygen because: 1

(a) it is warm **(b)** air is dense
(c) oxygen is heavy **(d)** air pressure and oxygen are low

8. Assertion (A): Thunderstorm clouds develop mostly in warm, moist air. Reason (R): Warm moist air rises quickly and cools to form clouds. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. A sudden fall in the barometer reading generally indicates: 1

(a) heavy snow only **(b)** an approaching storm
(c) a calm sea **(d)** clear weather

10. When air in a closed syringe is compressed to half its volume at constant temperature, its pressure: 1

(a) becomes zero **(b)** stays same **(c)** halves **(d)** doubles

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the eye of a cyclone dangerous when it passes, though it appears calm? 2

12. The cabin of an aeroplane has some pressure of 80 kPa while outside at height it is 25 kPa. Find the net outward pressure on the window of 0.1 square metre area and the force. 2

13. A tank has water 4 m deep. Find how the pressure at the base compares with that at 1 m depth. 2

14. Explain why the water in a dam stays contained in a thick curved wall. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the monsoon winds' reversal in direction between summer and winter in terms of land-sea heating. 3

16. Why are the walls of a submarine very strong? 3

17. Explain the changes in the weather before, during and after a thunderstorm. 3

OR

A weather report says the pressure in the centre of a cyclone is 95 kPa and the outside pressure is 101 kPa. Find the difference and explain its effect. 3

18. A force of 12 N acts on 3 square cm. Find the pressure in N per square cm and in Pa (1 square cm = 0.0001 square m). 3

OR

Why must we avoid using electrical appliances and a landline phone during lightning? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss how knowledge of pressure and winds is used in forecasting and managing cyclones and floods in India, with at least five measures. 5

OR

Analyse the statement: 'Air is invisible but exerts pressure'. Give four proofs from everyday life or experiment. 5

20. Explain why wind speeds increase and why the sea breeze moves inland during the day and compare with the land breeze at night, using pressure and temperature differences. 5

OR

Plan a school-level campaign to prepare students and families for storm season, listing five activities. 5

END OF QUESTION PAPER — Total: 40 Marks